

3.1.4 The use of finishes

Fabric finishes

Content	Potential links to maths and science
The effects of finishes and the reasons they are needed in relation to: • the fibre/fabric properties • end use of the product.	

Mechanical finishes

Content	Potential links to maths and science
A range of mechanical finishes, including: • brushing/raising • calendaring • embossing • heat setting using thermoplastic fibres to give permanent pleats or crinkles and make fabric non-crease and non-shrink • stone and sand washing.	Understand how the physical characteristics of fabrics can be modified by using mechanical finishes, eg trapped air acts as an insulator, air supports combustion, smooth fabric reflect light better than those with texture.

Chemical finishes

Content	Potential links to maths and science
A range of chemical finishes, including: • mercerisation • flame retardancy, eg Proban® and Pyrovatex® used on cotton fabrics • water resistance • non-iron/crease resistance, eg Teflon • shrink resistance and anti felting • moth proofing • anti-pilling • hygienic (sanitised). Detailed knowledge of the chemicals involved and methods of application is not expected.	Understand how the physical characteristics of fabrics can be modified by applying chemical finishes.

Laminating as a finishing process

Content	Potential links to maths and science
That a laminated fabric is a membrane system made up of two or more layers, and the principles behind how laminated fabrics work. Students should be able to describe a range of laminated fabrics, including: • Gore-Tex® • Sympatex® • Bonded fabrics, eg fake leather backed by a woven fabric to give stability.	